Supplier Reliability & Delivery Delay Report



Project Overview

Late deliveries quietly raise operating costs, frustrate customers, and strain supplier relationships. This SQL project analyzes order and seller data to show which suppliers deliver reliably and which do not.

Using nine relational tables, it compares each order's actual delivery date with its estimated date. It measures late-delivery counts, on-time and late rates, and average days late for every seller, then breaks delays down by product category, month and year, and seller state.

The result is clear evidence that businesses can use to monitor supplier performance, prioritize reliable sellers, and make better purchasing decisions.

What the analysis delivers



Seller reliability rankings



Delay severity and trends

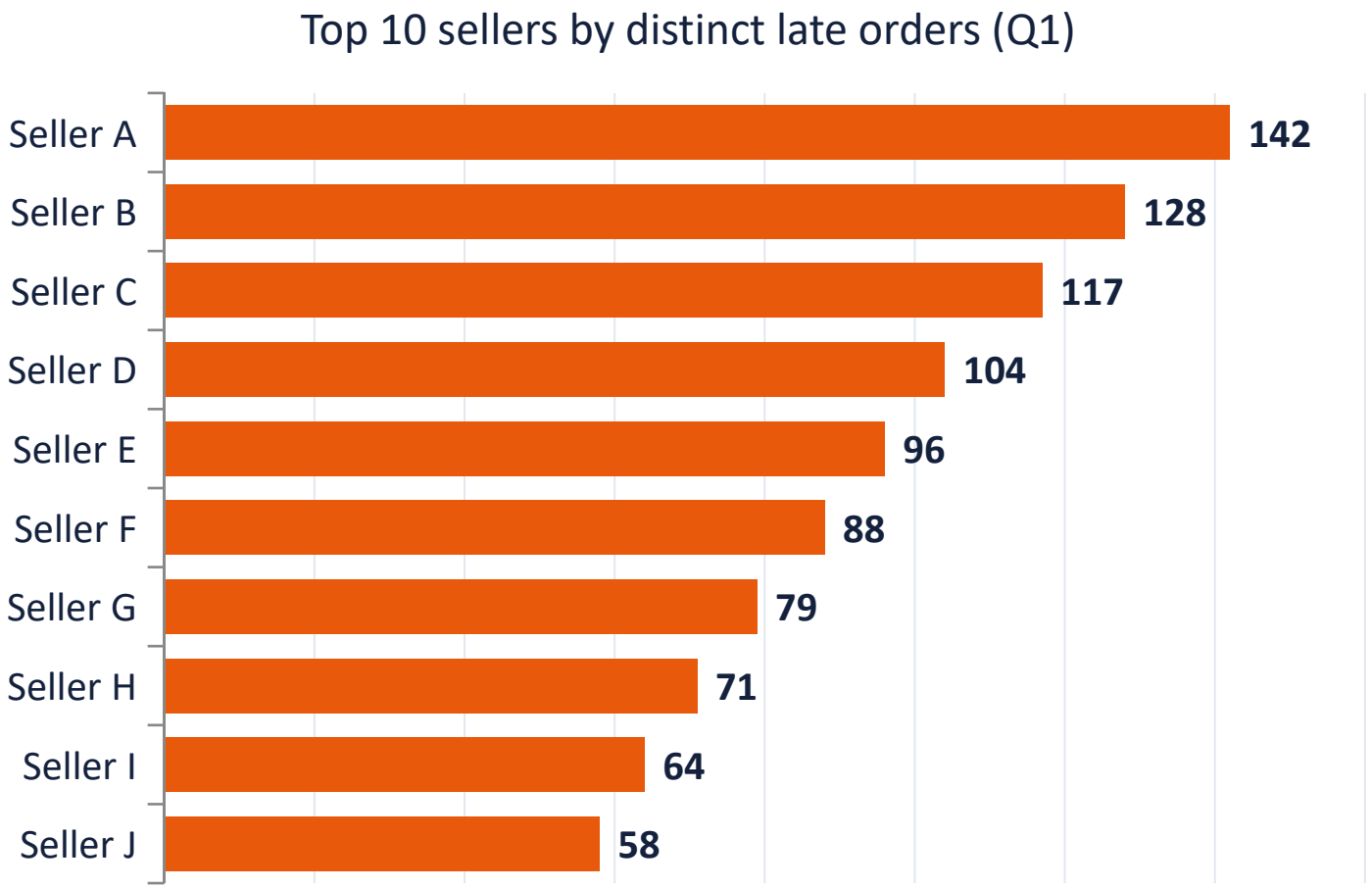


Category and regional hot spots



Purchasing recommendations

Sellers with Most Late Deliveries

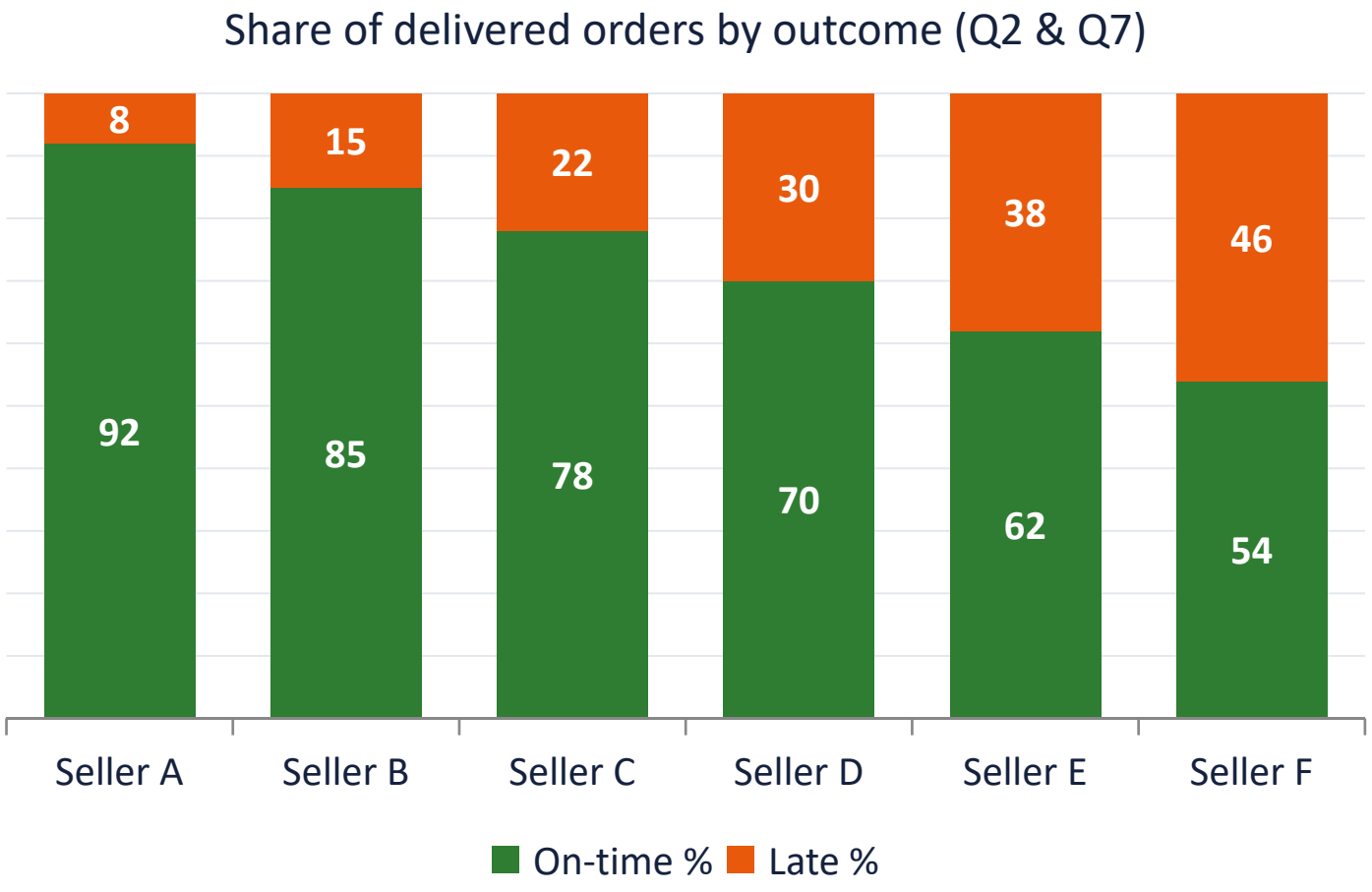


What it shows

- Ranks sellers by the number of distinct delivered orders that arrived after the estimated date.
- Highlights where delays are concentrated.
- High counts can reflect high volume, so read alongside late-delivery rate (Q7).

Illustrative sample data – replace with the output of your SQL query.

On-Time vs Late Delivery Rate

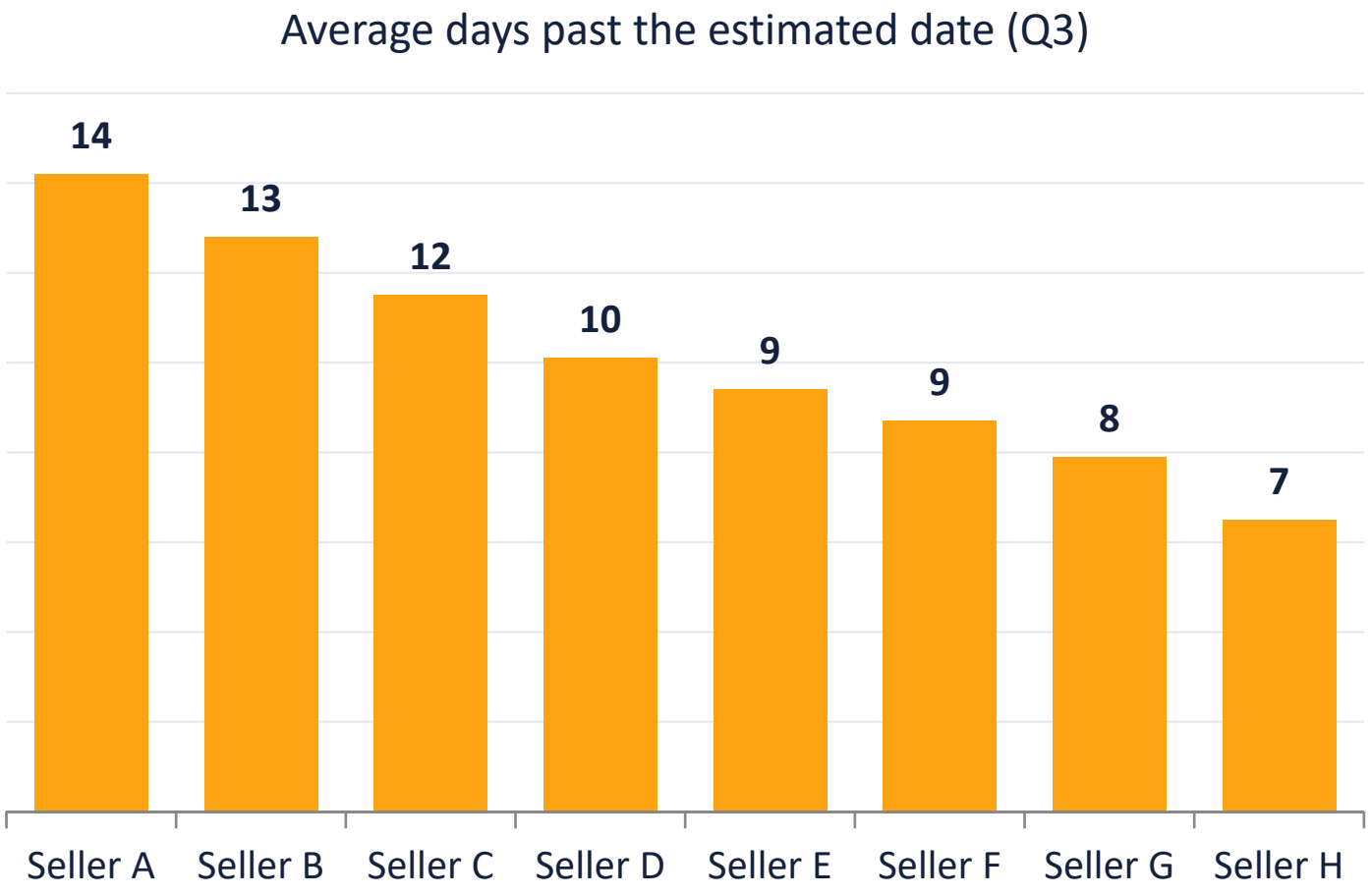


What it shows

- On-time rate (Q2) and late rate (Q7) are two sides of the same measure.
- Rates normalise for volume, so small and large sellers can be compared fairly.
- Sellers with low on-time rates are the first candidates for review.

Illustrative sample data – replace with the output of your SQL query.

Average Days Late by Seller

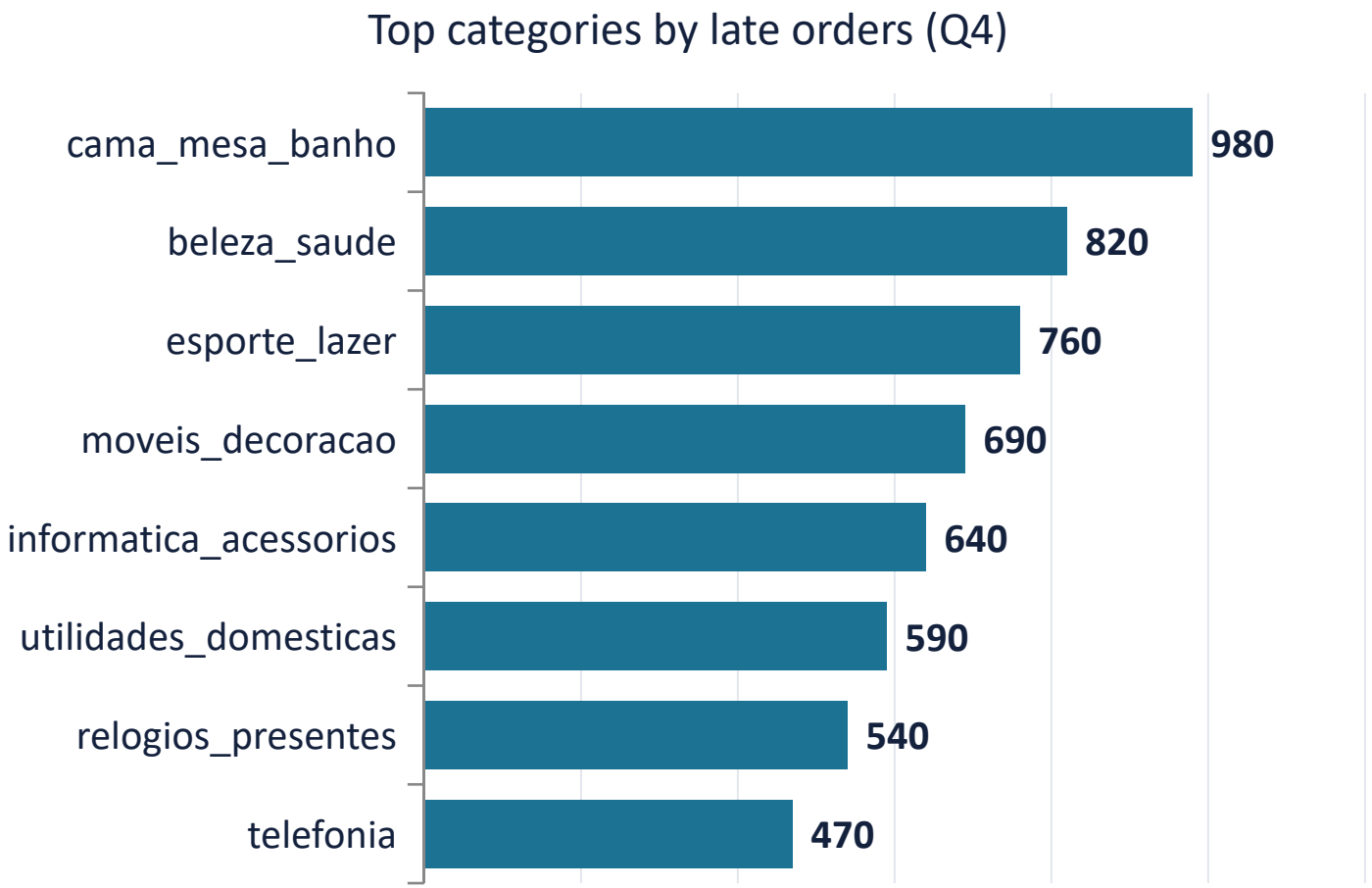


What it shows

- Measures the severity of delays, not just how often they happen.
- A seller can be late rarely but very late, which hurts customers more.
- Use it to separate minor slippage from serious failures.

Illustrative sample data – replace with the output of your SQL query.

Late Deliveries by Product Category

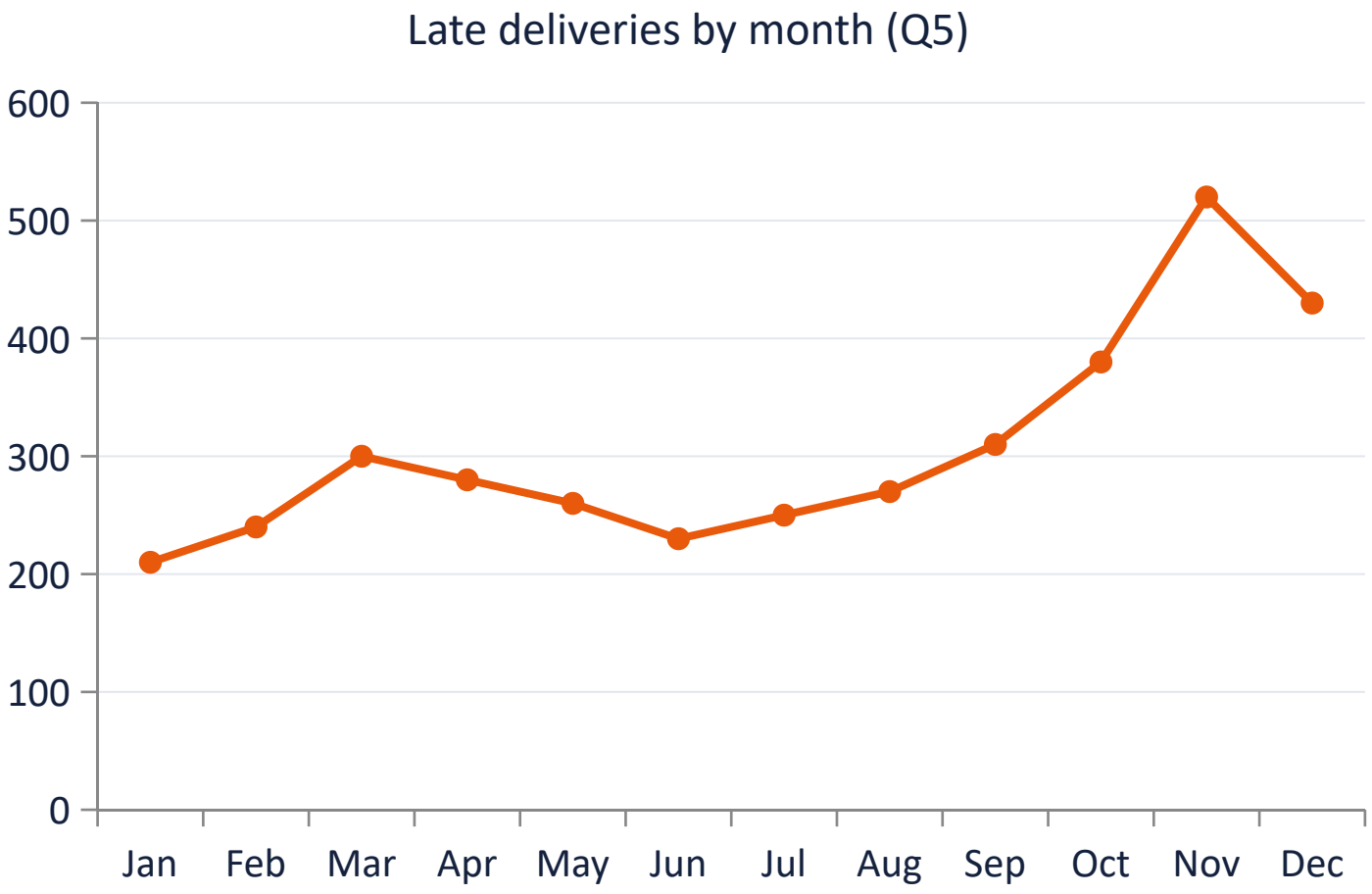


What it shows

- Shows which product categories experience the most delayed orders.
- Bulky or fragile categories often depend on slower logistics.
- Targets supply chain improvements where they matter most.

Illustrative sample data – replace with the output of your SQL query.

Late Deliveries Over Time

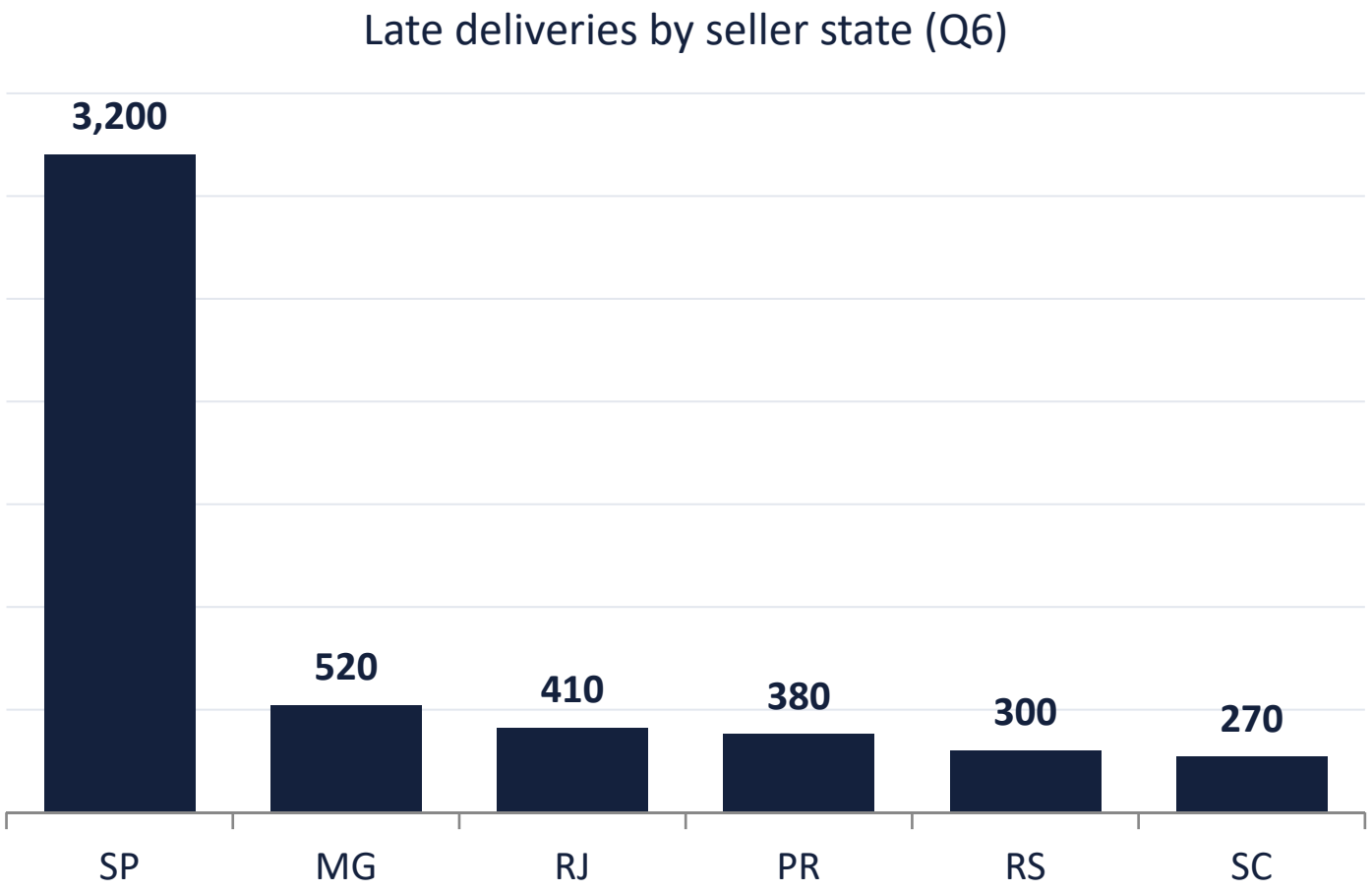


What it shows

- Reveals monthly and yearly delay trends by delivery date.
- Peaks usually line up with seasonal demand, such as holiday sales.
- Helps plan capacity and set realistic estimates before busy periods.

Illustrative sample data – replace with the output of your SQL query.

Late Deliveries by Seller State



What it shows

- Identifies the geographic areas where delays cluster.
- States with many sellers will naturally show higher counts, so compare rates too.
- Guides regional logistics and carrier planning.

Illustrative sample data – replace with the output of your SQL query.

Recommendations

Businesses should make supplier reliability a measured, regularly reviewed part of every purchasing decision. Evaluate suppliers on a monthly scorecard of on-time rate, late rate, and average days late, and direct more volume to consistently reliable sellers. Set improvement targets for chronic late sellers, strengthen coordination between sellers, carriers, and regional logistics, and use seasonal, category, and state patterns to prepare for peak periods.

Evaluate

Monthly scorecards of on-time rate, late rate, and days late.

Prioritize

Shift purchasing volume toward consistently reliable sellers.

Prevent

Fix delay drivers before peak periods, by category and state.

Conclusion

This project shows how SQL analysis of order and seller data turns late deliveries from a vague frustration into measurable evidence. By comparing actual and estimated delivery dates, it reveals which sellers, product categories, regions, and months drive delays. Acting on these insights helps businesses choose reliable suppliers, reduce operational costs, improve delivery efficiency, and strengthen supplier relationships and customer satisfaction.

